



FALL 2025

Course Title: Entrepreneurship

Course Code: CS-363

Assignment 1

Name of Course Instructor: Dr. Muhammad Faisal Rasheed		
Semester: eighth	Program: BSCS	Date: October 16 th 2025
Submission Date: October 26 th 2025	Maximum Marks: 30 Marks	
Program Objective: PO1, PO2	Course Objective: C2, C2, C6	GA: GA1, GA3, GA4
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Question No 1: As part of your university's Smart Campus project, you are invited to identify a real problem you face in your daily campus life (Campus +Hostel) and propose a technology-based solution. Using the Three Ways to Identify an Opportunity framework (Pain Points, Aspirations, and Trends), describe one specific issue that affects you or your peers, explain why it's worth solving, and suggest how your student software team could use technology to improve the experience.

Question No 2: Choose any popular computer science application or digital device that you use regularly, such as a mobile app, web platform, operating system, or smart gadget, and reimagine it using all seven SCAMPER techniques. Your task is to creatively redesign or enhance the chosen product by applying each SCAMPER step to propose new features, improvements, or use cases. Focus on solving real problems, improving user experience, and introducing innovative ideas that reflect your own thinking without relying on external help or AI tools.

Question No 3: Imagine you are planning to start your own tech-based business that addresses a real computing or AI-related problem faced by customers or the public. The problem could relate to data security for everyday users, digital accessibility, service automation, personalized learning, or efficient online transactions. Describe your business idea, explain what specific problem it solves for customers or society (**choose wisely, it will lead to subsequent assessments.**)

Guidelines:

1. 7-8 page (apart from cover page). The assignment will not be accepted without this cover page.
2. Headings should be handwritten, **and good presentations will be rewarded with good scores.**
3. **Plagiarized/copied/ LLM generated content will be marked ZERO.**
4. A soft copy (Pdf version) should be submitted via **QOBE** with the file name **"AI_last two digits your reg_ your first name_CS363."**
5. Font Size: Times New Roman, spacing, Headings and font must be consistent. **Use Bullets, headings, and graphics where required.**
6. Proper **Headings and good presentations will be rewarded with good scores.**
7. This is **individual assignment** and Failure to meet the guidelines leads to **zero marks** in this assessment.

QUESTION 1: SMART CAMPUS OPPORTUNITY IDENTIFICATION

Problem Identification: Professor Availability Tracking System

Pain Point Analysis

In our daily life on campus, students frequently face a significant challenge: professors and faculty members are not consistently available in their offices. This creates substantial friction in the academic workflow. When students have important work, coursework issues, or project discussions with their professors, they often must:

- Make multiple trips to the professor's office only to find them unavailable.
- Spend considerable time searching across the campus trying to locate their supervisor.
- Miss critical deadlines because they cannot access guidance when needed.
- Experience frustration and wasted effort due to unpredictable professor schedules.

This pain directly affects our current cohort at Namal University, particularly during intensive periods like Final Year Project (FYP) development. We consistently encounter situations where students need urgent feedback or guidance but cannot determine whether a professor is available without making the journey to their office.

Aspirations & Desired State

The ideal scenario would be a system where:

- Students can check professor availability in real-time from anywhere on campus.
- Professors can maintain their schedules transparently.
- Communication channels are streamlined and efficient.
- Students can schedule brief consultations without wasting time.
- Faculty can manage their office hours more effectively.
- There is a centralized hub for all academic guidance requests.

Market Trends Supporting This Solution

- **Smart Campus Initiative:** Universities globally are adopting IoT and digital solutions for campus management.
- **Location-Based Services:** Mobile applications increasingly leverage real-time location data.
- **Meeting Management Tools:** Tools like Google Calendar, Calendly, and Microsoft Bookings show increasing adoption in educational settings.
- **Campus Digitalization:** Post-pandemic, educational institutions are investing heavily in digital infrastructure.
- **Mobile-First Solutions:** Students prefer mobile-based solutions for accessing campus services.

Proposed Technology-Based Solution: Professora - Professor Availability Platform

System Overview

Professora is a mobile and web application that provides real-time professor availability tracking, integrated scheduling, and direct communication channels for academic consultations.

Core Features

1. Real-Time Availability Status

- Professors can update their status (In Office, In Class, In Meeting, Out of Campus, Office Hours)
- Color-coded indicators: Green (Available), Yellow (Busy/Office Hours), Red (Unavailable)
- Integration with university calendar systems to automatically update status based on scheduled activities.
- Optional QR code outside professor offices that displays real-time status.

2. Smart Scheduling System

- Students can view professor's office hours and available slots.

- One-click booking for consultation appointments (15, 30, or 45-minute slots)
- Automatic calendar invitations sent to both student and professor.
- Reminder notifications 0.5 hours before consultation

3. Communication Hub

- Integrated messaging for quick clarifications
- Video call capability for remote consultations
- File sharing for project discussions.
- Message history and documentation of academic interactions.

4. Analytics Dashboard (For Administrators)

- Faculty utilization of metrics
- Student feedback on consultation quality
- Peak demand times analysis
- Capacity planning insights

Why This Solution is Worth Solving

1. **Time Efficiency:** Reduces wasted trips and search time by 80%.
2. **Academic Quality:** Ensures students get timely guidance, improving project outcomes.
3. **Scalability:** Can be implemented across all Namal University departments
4. **Data-Driven:** Provides insights into improving academic support systems.

Technology Implementation Strategy

Technology Stack:

- **Frontend:** React Native (cross-platform mobile), React.js (web)
- **Backend:** Python FastAPI or Node.js Express
- **Real-Time Features:** WebSocket for live status updates
- **Calendar Integration:** Google Calendar API, Microsoft Graph API
- **Notifications:** Firebase Cloud Messaging, OneSignal
- **Database:** PostgreSQL for structured data, Redis for real-time status cache

Phase 1 (Development by Student Software Team):

1. Design user interface mockups and conduct feedback sessions with professors and students.
2. Develop MVP (Minimum Viable Product) with core availability tracking and scheduling.
3. Integrate with existing university calendar system.
4. Implement mobile app prototype and evaluate with pilot group.
5. Deploy web version for broader testing.

Phase 2 (Scaling & Enhancement):

1. Add location-based services (campus mapping integration)
2. Implement AI-based consultation recommendations.
3. Integrate with other campus systems (library, lab facilities)
4. Add analytics and reporting capabilities.
5. Campus-wide rollout and user training

Expected Impact:

- 70%+ adoption rate among students within first semester
- 25+ hours/month saved per professor through streamlined consultations.
- Improved student satisfaction scores by 40% - 50%
- Measurable improvement in project completion rates and academic performance

QUESTION 2: GITHUB PLATFORM SCAMPER ANALYSIS

GitHub is a critical collaboration platform for our development work and portfolio building. Below is a comprehensive SCAMPER analysis reimagining GitHub with innovative enhancements:

S - SUBSTITUTE: Replace Traditional Features

Current Feature: Static README files

Substitution Proposal: Interactive executable documentation

- README transformed into interactive Jupyter notebooks
- Live code examples that execute directly in the browser
- Step-by-step visual tutorials with animated demonstrations
- API documentation with built-in request/response testing

C - COMBINE: Merge Different Elements

Combination 1: GitHub + Project Management Tools

- Integrate Kanban boards directly within repository.
- Combine issues with time tracking and resource allocation.
- Automated sprint planning based on GitHub issue analysis.
- Burndown charts automatically generated from PR merge patterns.

A - ADAPT: Adjust for Specific Contexts

Adaptation 1: GitHub for Non-Technical Creators

- Simplified interface hiding complex Git terminology.
- Visual drag-and-drop file management.
- One-click deployment without CLI knowledge
- Automatic Git commands executed behind the scenes.

M - MODIFY: Change & Enhance Existing Elements

Modification 1: Enhanced Code Review System

- AI-powered code review that suggests improvements before human review.
- Automated detection of security vulnerabilities, performance issues, design patterns
- Visual diff improvements showing tree-structure changes for complex refactors.
- ML-based suggestion confidence scores

P - PUT TO ANOTHER USE: Alternative Applications

New Use Case 1: GitHub as Academic Research Hub

- Version control for research datasets, experiments, and papers
- Reproducible science through tracked environments and dependencies.
- Collaborative research paper writing with version history
- Preprint publishing directly through GitHub.
- Citation tracking and impact metrics.

E - ELIMINATE: Remove Unnecessary Elements

Elimination 1: Remove Redundant Interfaces

- Consolidate Gists, Wikis, and Discussions into unified content system.
- Single unified authentication instead of multiple credential options
- One notification center instead of scattered notifications
- Eliminate the required workflow; allow direct collaboration on branches.

R - REVERSE: Flip the Approach

Reversal 1: Decentralized GitHub Nodes

- Instead of centralized cloud platform, create federated GitHub instances.
- Local organizations run their own GitHub servers.
- Seamless synchronization between nodes for collaboration
- Privacy and data control maintained locally.
- Resilience against platform outages

QUESTION 3: TECH-BASED BUSINESS STARTUP IDEA

Project Title: CodeOptimize - AI-Powered Code Generation and Optimization Platform

Business Overview

Based on our Final Year Project (FYP) proposal, CodeOptimize is a technology-enabled platform addressing the critical gap in current AI-assisted code generation tools. While tools like GitHub Copilot and ChatGPT excel at generating functional code, they often produce computationally inefficient solutions.

The Real Problem Being Solved

Problem Statement:

Modern AI code generation tools prioritize functionality over efficiency. They produce code that "works" but often suffers from:

- High time complexity $O(n^3)$ instead of optimal $O(n \log n)$
- Excessive memory consumption
- Redundant operations and anti-patterns
- Poor scalability for enterprise applications
- Increased infrastructure costs for companies

Real-World Impact:

- A software company with 100,000+ lines of AI-generated code paying 40% more in cloud computing costs due to inefficient algorithms.
- Startup developers are unaware of optimization opportunities, leading to technical debt accumulation.
- Machine learning engineers are unable to generate optimized code for resource-constrained edge devices.
- Competitive disadvantage when infrastructure costs exceed optimized competitors by 30-50%

Target Customer Pain:

- Junior developers: Do not know algorithmic optimization best practices.
- Startups: Cannot afford DevOps specialists for code review
- Enterprise teams: Need systematic code quality assurance.
- Machine learning practitioners: Require optimized inference code for deployment.

The CodeOptimize Solution

What Problem It Solves for Customers

For Individual Developers:

- Generate code is both functional AND mathematically optimized.
- Learning optimization best practices directly from the system.
- Reduce time spent on code reviews focused on efficiency.
- Build stronger portfolios with optimized code samples.

For Startups:

- Reduce cloud infrastructure costs by 30-40% through optimized code.
- Accelerate development without sacrificing performance.

- Eliminate need for expensive code optimization specialists.
- Scale products efficiently from day one.

For Enterprises:

- Enforce code quality standards through AI-powered optimization.
- Reduce technical debt accumulation.
- Ensure compliance with performance requirements.
- Make informed decisions about optimization ROI.

Product Description

Core Offering: CodeOptimize Platform

A conversational AI assistant that generates optimized code by combining:

- 1. NLP-Based Intent Understanding**
 - Interprets complex code generation requests.
 - Distinguishes between optimization vs. generation tasks.
 - Extracts performance requirements and constraints
 - Multi-language support (Python, JavaScript, Java, C++, Go, Rust)
- 2. Context-Aware Code Generation**
 - Generates code using fine-tuned LLMs (GPT-4/Claude)
 - Retrieves relevant patterns from curated best practices database.
 - Incorporates domain-specific knowledge (algorithms, data structures)
 - Provides multiple solution options with trade-off analysis.
- 3. Mathematical Optimization Engine**
 - Analyzes generated code for time/space complexity.
 - Suggests algorithmic improvements with formal proofs.
 - Applies optimization transformations automatically.
 - Shows before/after performance metrics (Big-O analysis)
- 4. Code Analysis & Refactoring**
 - Detects anti-patterns and inefficiencies.
 - Recommend design pattern applications.
 - Identify unnecessary computations.
 - Suggests data structure optimizations.

Revenue Model

Tiered SaaS Model:

- 1. Starter Plan - \$19/month**
 - One hundred code optimization requests/month
 - Support for three programming languages
 - Basic complexity analysis
 - Community forum access
- 2. Professional Plan - \$79/month**
 - Unlimited optimization requests
 - Support for 6+ programming languages
 - Advanced pattern recommendations
 - API access for IDE integration
 - Priority email support
 - Private knowledge base
- 3. Enterprise Plan - Custom**

- On-premises deployment option
- Custom optimization models for proprietary patterns
- Resolute account manager
- Integration with existing CI/CD pipelines
- Team licenses (5-50 developers)
- Advanced reporting and analytics
- SLA guarantees

4. Freemium Features

- Five free optimization requests/month
- Single language support
- Upgrade prompts within tool.
- Community library access

Projected Revenue (Year 1):

- 500 Starter users $\times \$19 \times 12 = \$114,000$
- 200 Professional users $\times \$79 \times 12 = \$189,600$
- 5 Enterprise contracts $\times \$2,000/\text{month} \times 12 = \$120,000$
- **Total Year 1 Revenue: ~\$423,600**

Market Opportunity

Total Addressable Market (TAM):

- 4.4 million software developers globally
- Fifty percent use AI code generation tools
- Thirty percent are willing to pay for optimization.
- Addressable market: ~660,000 potential customers
- At average \$50/month: \$396 million TAM

Serviceable Market (SAM):

- Initial focus: US and European markets
- 200,000 active AI code generation users
- Conservative adoption: 5% = 10,000 paying users
- 5-year SAM: \$6 million

Market Trends Supporting Growth:

- AI code generation adoption increased 45% YoY.
- Cloud computing costs are rising by 20%+ annually.
- Developer shortage driving tool adoption.
- Enterprise focus on technical debt reduction

Competitive Analysis

Feature	CodeOptimize	GitHub Copilot	ChatGPT	Cursor AI
Code Generation	✓	✓	✓	✓
Optimization Focus	✓	✗	✗	✗
Mathematical	✓	✗	✗	✗

Feature	CodeOptimize	GitHub Copilot	ChatGPT	Cursor AI
Analysis				
Multi-Turn Context	✓	Limited	✓	✓
Best Practices RAG	✓	✗	✗	Limited
Performance Metrics	✓	✗	✗	✗
SaaS Pricing	✓	Enterprise	✓	Limited

Competitive Advantages:

- 1. Only platform specializing in optimization-first code generation.
- 2. Mathematical rigor with formal complexity analysis
- 3. Curated best practices knowledge base.
- 4. Educational value for developer upskilling
- 5. Measurable ROI through cost reduction

Business Implementation Strategy

Phase 1: MVP Launch (Months 1-4)

- Develop MVP supporting Python and JavaScript
- Five hundred beta testers from developer communities
- Validation of optimization claims through benchmarking.
- Initial customer feedback integration

Phase 2: Market Entry (Months 5-8)

- Launch public SaaS platform.
- Freemium model to drive adoption.
- Partnerships with coding bootcamps and universities
- Content marketing highlighting optimization benefits.

Phase 3: Scaling (Months 9-12)

- Expand to 6+ programming languages.
- Enterprise sales team formation
- Integration partnerships (VS Code, IntelliJ, GitHub)
- API product launch for IDE integration

Phase 4: Market Leadership (Year 2)

- Enterprise contracts with Fortune 500 companies
- International expansion (EU, Asia-Pacific)
- Advanced features: Custom model training, AI-powered refactoring
- Strategic acquisition targets or partnerships

Why This Business Idea is Worth Pursuing

1. **Massive Market Need:** AI-generated code quality is a recognized industry pain point.
2. **Clear Value Proposition:** Measurable ROI through cost reduction (30-40% savings)
3. **Defensible Technology:** Proprietary optimization engine and knowledge base
4. **Scalable Model:** SaaS platform with minimal marginal costs
5. **Large TAM:** Developer tools market worth billions
6. **Trending Topic:** AI and code optimization are extremely hot in 2024-2025.
7. **Alignment with FYP:** Directly leverages our Final Year Project work.
8. **Team Expertise:** Our project team has deep knowledge of AI/ML, optimization, and code analysis.

Societal Impact

Beyond commercial success, CodeOptimize creates positive societal outcomes:

- **Environmental:** Reduced computing resources = lower carbon footprint
- **Economics:** Smaller companies can compete with large enterprises efficiently
- **Educational:** Developers learn optimization best practices at scale
- **Accessibility:** Making software development more efficient and affordable global

